The Hong Kong Polytechnic University
Department of Building Environment and Energy Engineering

BSE3124 Electricity and Buildings **2026/27 Semester 1**

Assignment – Answer Sheets

Group number: 1

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Task 1 – Loading estimation and transformer room location

Estimation of the building electrical load using CLP COP215 and justifications for transformer room and main LV switch room locations:

Basis and assumptions. The assessment covers ten identical office floors, each with usable areas assumed to be 309, 309 and 410 m^2. CLP COP215 Rev 07 forms the load-assessment basis. The 1000 kW in the question is cooling output, not electrical demand. Additionally, the calculation assumes packaged air-cooled AC, PF 0.90, five lifts and equal allocation to two supply zones.

$$
\begin{aligned}
A &= 10(309 + 309 + 410) = 10\,280\ \mathrm{m}^2, \\
S_{\text{office}} &= 10\,280(0.16) = 1644.8\ \mathrm{kVA}, \\
P_{\text{AC}} &= (1000/3.51685)(1.85) = 526.039\ \mathrm{kW}, \\
S_{\text{AC}} &= 526.039/0.90 = 584.488\ \mathrm{kVA}, \\
S_{\text{public}} &= 5(40) = 200\ \mathrm{kVA}, \\
S_{\text{design}} &= 1.10(1644.8 + 584.488 + 200) = 2672.216\ \mathrm{kVA}.
\end{aligned}
$$

The 40 kVA/lift allowance is an aggregate public-services reference, not a lift motor rating. A public-services overlap allowance of 0–200 kVA gives 2452.216–2672.216 kVA; the upper bound is adopted. The references are COP215 3.1.7, 3.3.1 and Appendices 2/7.

Selection. The proposal uses two 1500 kVA transformers to provide 3000 kVA. Each serves five floors and half the plant/public allowance: 1336.108 kVA per section. Full-load N–1 is not provided; a failed transformer requires load shedding. Unauthorised parallel operation must be prevented.

Room location. The lower-left ground-floor perimeter accommodates the proposed 12430×5450 mm transformer room and adjacent 12430×3500 mm main LV room (LV depth assumed). This allows external delivery/access, short LV interfaces, separation from the central lift/stair/AHU core and independent customer access.

Access provisions include 3 m delivery access within 20 m of unloading. Its planning conditions include COP101 v16 4.3 clearance envelopes, 2-hour partitions, sealed penetrations and a raised floor level. Door sweeps require sequential use. Louvres provide 1.20 m^2 free area at the assumed 50% free ratio. Equipment/ventilation and 10.03/10.31 m LV interfaces are detailed in the continuation and ground-floor drawings.

Task 1 – Loading estimation and transformer room location (continued)

Transformer and main LV room proposal

The proposal starts with two 1500 kVA transformers serving separate LV sections, providing 3000 kVA in total and a margin of 327.784 kVA. Floors 1–5 are allocated to section A and floors 6–10 to section B, with each section also receiving half of the central plant/public allowance. Under this assumed equal split, each section carries 1336.108 kVA, corresponding to about 2030 A at 380 V. However, full-load N–1 is not provided, so load shedding is required if one unit is lost. The two LV sections operate independently; no bus coupler is provided.

Room placement follows the lower-left perimeter of the supplied plan, with a 12430 by 5450 mm substation and an adjacent 12430 by 3500 mm LV room, whose depth is assumed. This location allows external plant access and short LV interfaces, while providing separation from the lift/stair/AHU core and independent LV access. Additionally, a 3 m clear delivery route within 20 m of unloading is proposed, subject to confirmation of the external site conditions.

The plantroom drawing shows HV-01 as a compartmented switchboard and TX-A/TX-B as transformer plan symbols with HV/LV terminal positions. Adopted overall dimensions are 3000 by 1000 mm for HV-01 and 2200 by 1800 mm for each transformer, with six LV cabinets. These are design dimensions; internal equipment details are diagrammatic. Their clearance envelopes fit without overlap. COP101 v 16 clause 4.3 requires HV clearances of 1000 mm at the rear, 1800 mm at the front and 750 mm at the sides, together with 900 mm at transformer LV terminals and 750 mm elsewhere. Additionally, the proposal includes 2-hour partitions, sealed penetrations and an FFL 150 mm above external level. Meanwhile, the outward-opening 180-degree doors have shared sweeps and therefore require sequential use.

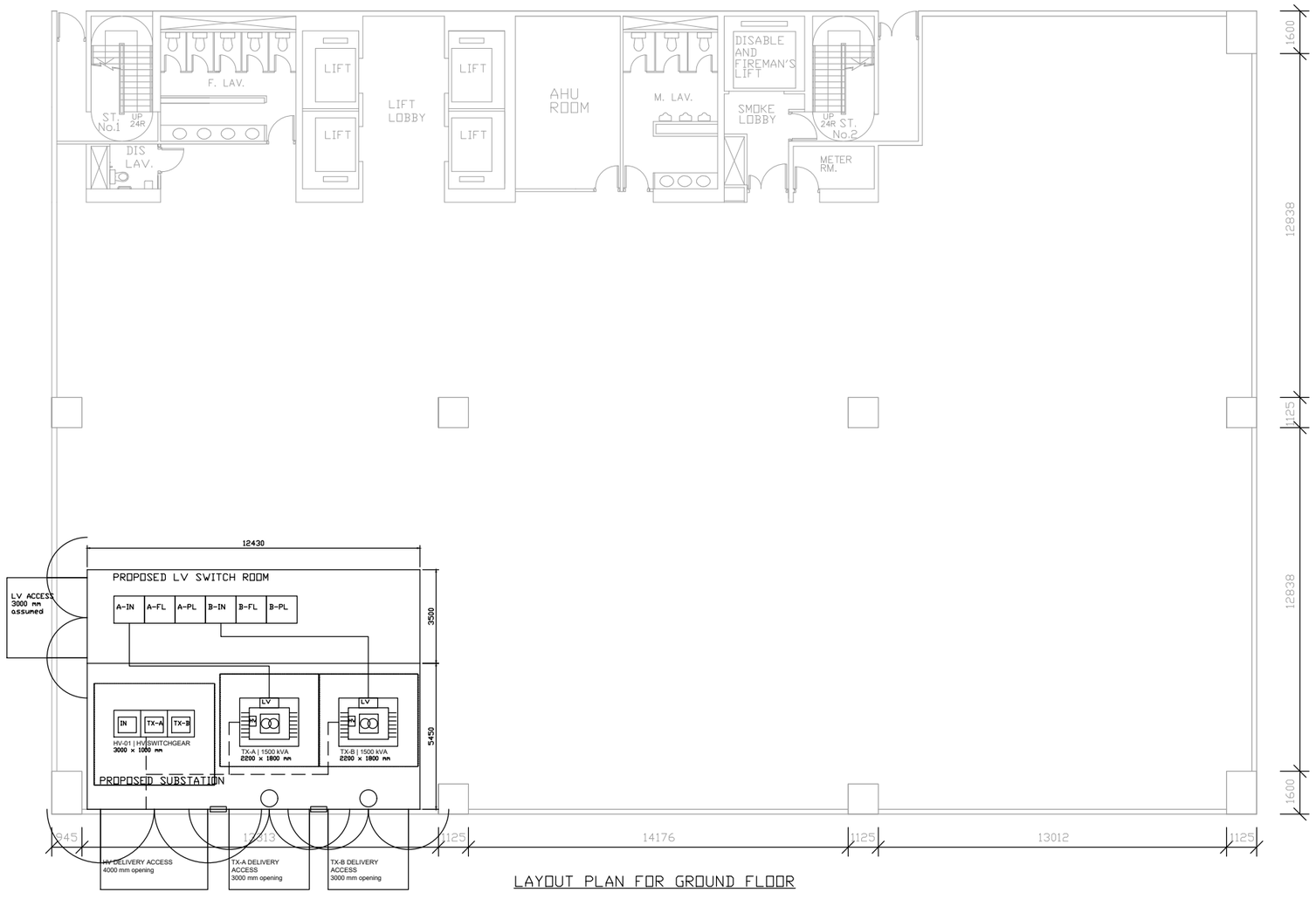
Ventilation is provided by two 600 by 2000 mm louvres, providing 1.20 m^2 at an assumed 50% free-area ratio, above clause 6.3's 1.12 m^2 reference. Thus, the certified free-area ratio must be at least 46.67%. Fan symbols are conditional provisions, not actual heat-loss calculations. Meanwhile, the LV interfaces, including the assumed tails, are 10.03/10.31 m, below the 20 m limit in 5.5.2.2. Actual equipment, structures, openings and ventilation must satisfy these declared conditions.

Main distribution schematic

The main schematic connects TX-A and TX-B through their respective ACB incomers to MSB-A and MSB-B. Separate rising supplies serve floors 1–5 and 6–10, with three tenants per floor. Each section also feeds its allocated central plant and public services. The Office 1 inset links floor metering to MS-O1, the selected feeder and DB-O1. The arrangement follows the transformer, switchboard and metering relationships in *Electrical Schematic 1*; the adopted loads and final-circuit ratings remain those calculated in Tasks 1 and 3.

TASK 1 | GROUND-FLOOR PLANTROOM LOCATION

A3 LANDSCAPE | SCALE 1:200 | DIMENSIONS IN mm | GROUND FLOOR



LAYOUT PLAN FOR GROUND FLOOR

PROPOSED LOCATION
Lower-left perimeter of supplied ground-floor plan.
Substation 12430 x 5450; adjacent LV room 12430 x 3500 assumed. See separate 1:75 room detail.

LOCATION REASONS
Direct external plant access is proposed, subject to site confirmation.
Adjacent LV room keeps transformer-to-incomer interfaces short.
Perimeter plantrooms separate supply equipment from the upper lift/stair/AHU core and remaining accommodation.
LV room has its own proposed west entrance.

LOAD ASSESSMENT
Offices: 1644.800 kVA.
AC: 1000 kW cooling -> 526.039 kW electrical -> 584.488 kVA at assumed PF 0.90.
Public services: 200 kVA aggregate allowance.
Two-zone conservative envelope: 2672.216 kVA.
Propose 2 x 1500 kVA, with equal zone allocation; no full-load N-1.

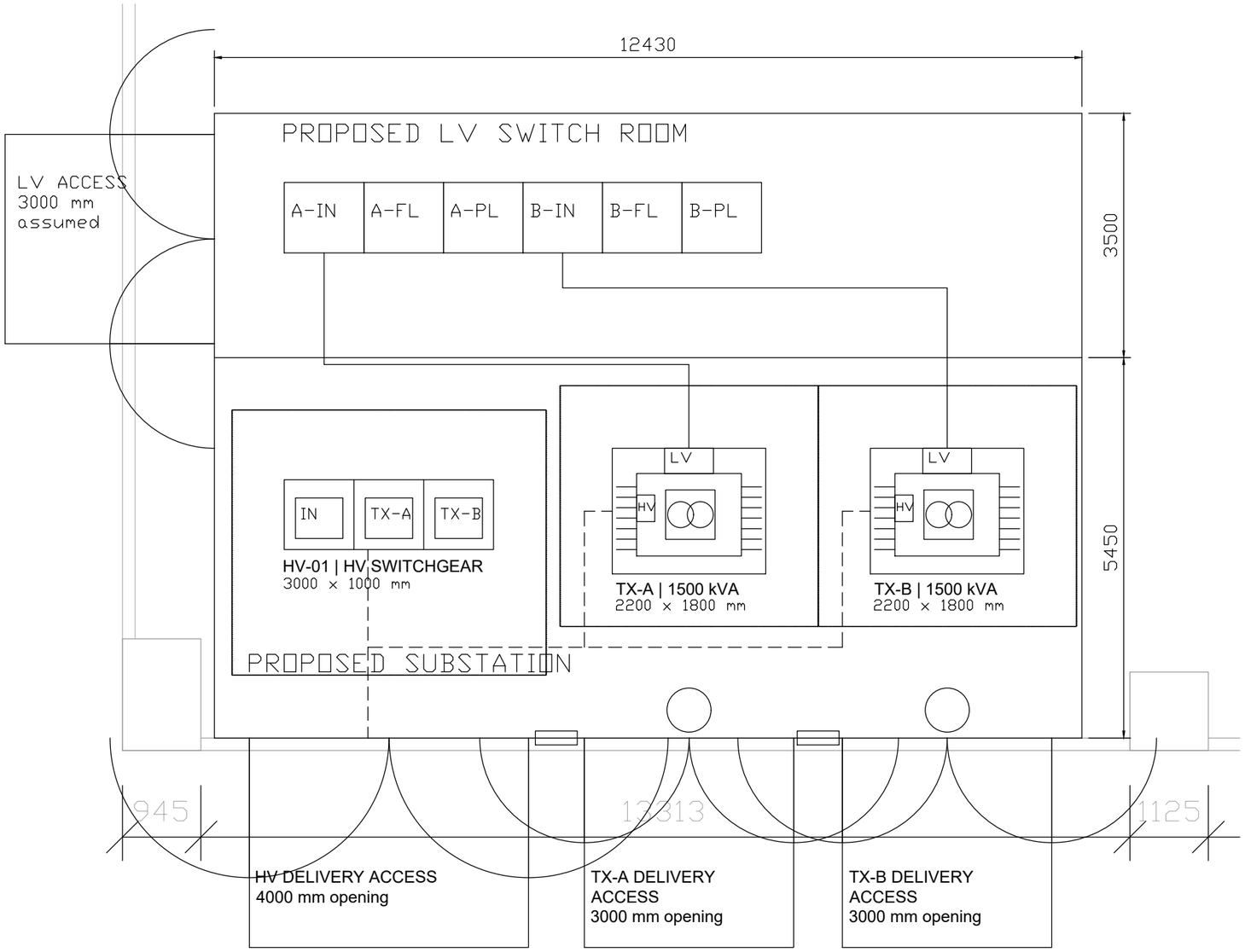
SITE CONDITIONS TO VERIFY
External apron, road and <=20m unloading route are not given by the source plan. Maintain >=3m clear delivery route.
Architectural wall openings, structural/flood/fire provisions, ventilation and supplier equipment approval require coordination.

GROUP 1 | TASK 1

Design basis: CLP COP215 Rev07; CLP COP101 v16.

TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

A3 LANDSCAPE | SCALE 1:75 | DIMENSIONS IN mm | PLANTROOM DETAIL



LINE KEY: dashed = HV trench; solid = LV route; chain = clearance. Equipment details are diagrammatic.

LV INTERFACES / HV ROUTES
LV trench: 1000 W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.
HV centre-lines: 1000 W at panels; 600 W branches; final 300 W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3

CAPACITY AND EQUIPMENT
TX-A / TX-B: 1500 kVA each. Total 3000 kVA; design load 2672.216 kVA. No full-load N-1.
HV-01: 3000 x 1000 mm; TX-A/B: 2200 x 1800 mm.
Cabinets: A/B-IN incomer; A/B-FL floor feeders; A/B-PL plant/public feeders.
Generic plan symbols; overall dimensions are adopted design assumptions. Internal details are diagrammatic.

CLEARANCES / BUILDING WORKS
HV rear 1000, front 1800, sides 750. TX LV terminals north: 900; other sides 750.
Proposed 2h FRR partitions; fire-seal penetrations. FFL +0.15 vs external +0.00 assumed.
Actual structures, wall thickness and equipment dimensions require coordination.

ACCESS AND DOORS
South external access assumed. Provide 3000-wide clear delivery route; unloading distance <=20 m to verify.
Plant doors: HV 4000 W; TX 3000 W each; 2800 H assumed. Outward opening, shown at 90 degrees.
Adjacent plant doors used sequentially; TX-A has wicket personnel access.
Independent LV entrance west.

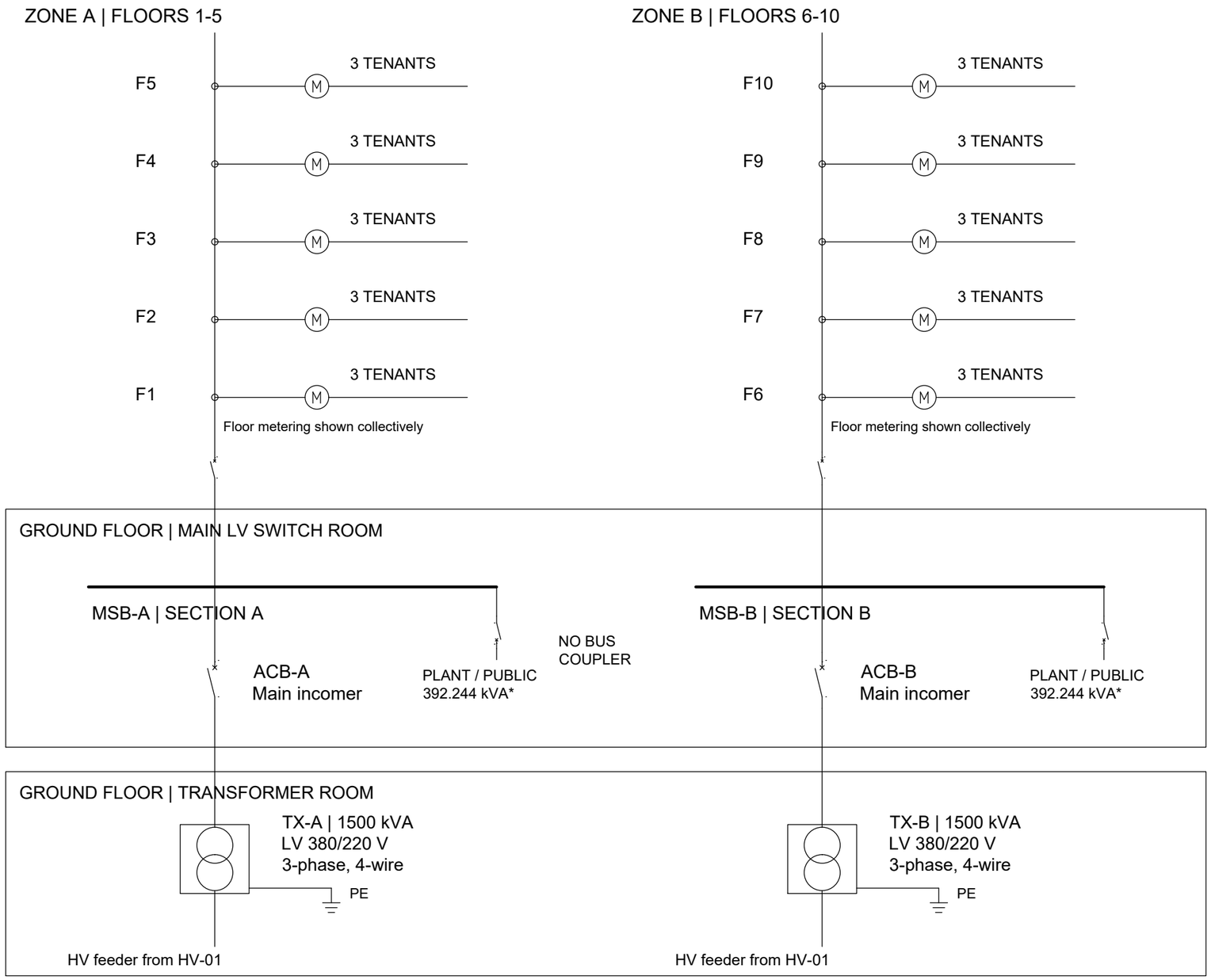
VENTILATION
2 low-level 600 W x 2000H inlet louvres: 2.4m2 gross x assumed 50% free area = 1.20m2 >=1.12m2. Certified free area >=46.67% required.
2 high-level 630-dia fan symbols are conditional references. Actual losses, fan duty and noise to confirm; outlet >=2.5m above street, away from people.

Basis: supplied COP215 Rev07; COP101 v16.
See the new main distribution schematic for TX-A/B, MSB-A/B and floor supply zones.

TASK 1 | MAIN ELECTRICAL DISTRIBUTION SCHEMATIC

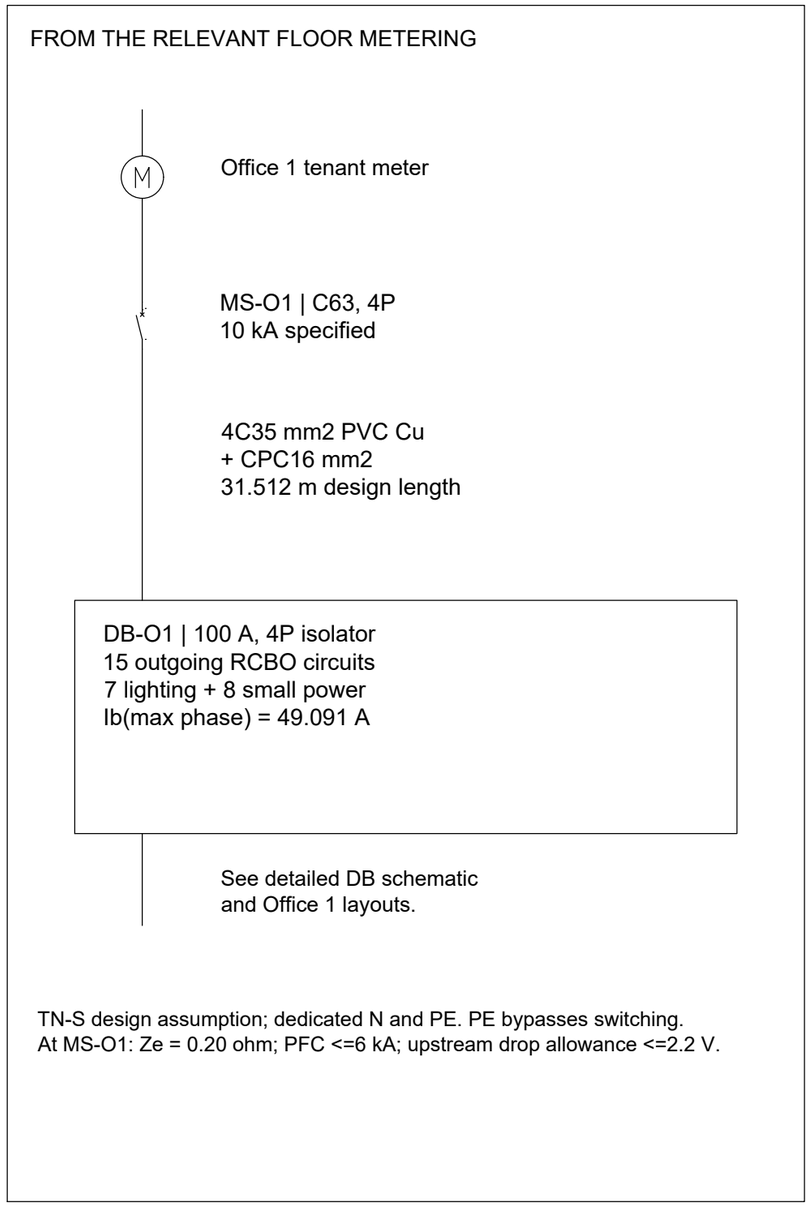
A3 LANDSCAPE | NOT TO SCALE | 2 x 1500 kVA | 380/220 V LV DESIGN BASIS

NORMAL SUPPLY: 2672.216 kVA total; two independent LV sections.
Equal allocation: 1336.108 kVA / approx. 2030 A per section.



* Per section: AC 292.244 + public services 100 = 392.244 kVA before the 1.1 zoning factor.

TYPICAL OFFICE 1 SUPPLY
Selected tenant detail; applicable to its floor supply zone.



TN-S design assumption; dedicated N and PE. PE bypasses switching.
At MS-O1: Ze = 0.20 ohm; PFC <=6 kA; upstream drop allowance <=2.2 V.

SYSTEM BASIS
HV supply voltage and upstream protection: utility interface. Capacities and LV voltage follow Task 1. Main ACB ratings/settings and fault duties require coordinated equipment selection. PE denotes equipment bonding; source earthing arrangement to be coordinated.

DESIGN LIMITS
No full-load N-1. Central plant/public load is split equally as a design assumption. This sheet shows normal supply; emergency systems are separate.

REFERENCE: Electrical Schematic 1 - transformer, main switchboard and floor-metering arrangement.
Read with the plantroom layout and Office 1 distribution-board schematic.

GROUP 1 | E-04 | REV 01 | 10 NOV 2026

Task 2 – Layout drawings and MCB board schematics

The office tenant chosen for layout design: Office 1.

Design considerations of circuit arrangement and protective device selection for small power and lighting layout:

Selected tenant: Office 1. The attached lighting/power plans, completed board template and detailed L/N/PE schematic present the design.

Lighting. Normal lighting uses 92 PIL PLATO 600 panels: sixty 22 W, twenty-eight 28 W and four 38 W, total 2256 W. Seven circuits LC-A–LC-G have 18/17/18/11/14/10/4 lamps, each protected by B10 Type A 30 mA 1P+N RCBO. Each has its own neutral and local manual control S-A–S-G. Different-phase switches use separate enclosures. Ceiling-grid and furniture relationships guide positions.

Lighting performance. Manufacturer LDT calculations give maintained mean direct illuminances of 592.9/546.3/588.5/662.6 lx for the open office/pantry/upper east/south east rooms at 1.9 m above the workplane and MF 0.80. Four 38 W panels in the south east room retain 589.2 lx at 2.2 m, providing a 17.8% margin above the 500 lx design target. The continuation gives the calculation method and room results.

Power. Small-power provision comprises nine four-socket floorboxes, three twin sockets and FS01. Every assembly has one upstream 13 A FCU; its outlets share that capacity. P1–P4 use 6 mm^2 A1 rings/B32, P5/P6/P8 use 10 mm^2 A2 radials/B32, and P7 uses a 4 mm^2 dedicated radial/B16. Each local fused spur is 4 mm^2, assumed 0.5 m. FCUs precede the spur cable. Additionally, the pantry circuits are separated from ordinary office circuits. FS01 reserves 13 A without inventing an appliance rating.

Protection and containment. Ring protection requires L/N/CPC continuity, with both ends returning to the same circuit under EMSD Table 6(1)'s A1 ring conditions. The lighting group, ring banks A/B, radial bank C, P7 and incoming feeder remain separate to preserve the stated grouping factors, as shown on the installation-detail sheet.

Supply and controls. The supply uses source C63 4P protection, 4C35 mm^2 + CPC16 and 100 A 4P DB isolation. Dedicated neutrals pass through their own RCBO; PE is unswitched. Accordingly, the selected apparatus must meet the specified 10 kA breaking capacity and total-energy/inrush limits. Landlord emergency lighting and exit signs use independent essential supplies.

Task 2 – Layout drawings and MCB board schematics (continued)

The office tenant chosen for layout design: Office 1.

Normal lighting and local controls

The lighting proposal starts with 92 PIL PLATO 600 MultiPower 3117955 panels on the 600 mm ceiling grid: sixty 22 W, twenty-eight 28 W and four 38 W, totalling 2256 W. Their nominal fluxes are 2843.45, 3555.30 and 4685.55 lm respectively. Separate circuits serve the enclosed rooms and distributed main-office zones, with a maintained mean illuminance target of 500 lx adopted for the design. With this arrangement, the connected lighting density is 7.301 W/m^2 using the adopted 309 m^2 office area.

Seven passive manual switches S-A–S-G provide local control of their corresponding circuits. These switches must have a minimum rating of 20 AX, an LED switching capacity of at least 600 W and make/break/inrush compatibility with the selected drivers. Since their phases differ, S-A/B/C use separate enclosures. No borrowed neutral is allowed. Each route passes through its switch and includes a 2 m DB rise plus 1.5 m down and 1.5 m back at the switch, giving an assumed vertical allowance of 5 m per lighting radial. Passive controls add no operating power.

Circuit	Lamps	W	Phase	RCBO	Control
LC-A	18	402	L1	B10	S-A
LC-B	17	392	L2	B10	S-B
LC-C	18	396	L3	B10	S-C
LC-D	11	242	L3	B10	S-D
LC-E	14	392	L2	B10	S-E
LC-F	10	280	L1	B10	S-F
LC-G	4	152	L3	B10	S-G

On the plan, the A/B/C numeric labels correspond to the LED IDs and circuits LC-A/B/C. Similarly, D01–D11 correspond to PN01–PN11, E82–E96 excluding E90 to the northeast LEDs, F01–F10 to UE01–UE10, and G01–G04 to SE01–SE04. Power settings follow the common device schedule. Every luminaire and control has one circuit assignment.

Lighting calculation

To assess lighting performance, the calculation uses manufacturer LDT intensities given in cd/klm, with each profile scaled by its selected flux: 2843.45 lm at 22 W, 3555.30 lm at 28 W and 4685.55 lm at 38 W. Illuminance is then calculated at each workplane grid cell, with spacing at most 0.2 m, as

$$E = 0.80 \sum_j \frac{I_j(C,\gamma)\cos\gamma}{d_j^2}.$$

A luminaire-to-workplane height of 1.9 m is adopted, corresponding to a 2.7 m ceiling and a 0.8 m workplane. Meanwhile, the maintenance factor of 0.80 assumes periodic cleaning and replacement. Room boundaries use the adopted plan envelopes; each room receives light from its own fittings. Only direct light is included in this calculation.

Task 2 – Layout drawings and MCB board schematics (continued)

Zone	Lamps	Area m^2	Mean lx	Min/mean
Open office	67	231.87	592.9	0.136
Pantry	11	29.54	546.3	0.390
Upper east room	10	33.68	588.5	0.267
South east room	4	13.07	662.6	0.175

On this basis, all four mean illuminances exceed the adopted 500 lx target. To provide mounting-height tolerance, the four south-east panels use the 38 W setting. At 2.2 m above the workplane, the respective means are 573.0, 505.8, 545.0 and 589.2 lx; the south-east room retains a 17.8% margin.

When interpreting these results, the minimum/mean ratios include perimeter cells. Overall, the assessment covers mean direct illuminance; a detailed task-area check would also account for furniture, reflections and glare.

Small power and fixed equipment

Small-power provision includes nine four-socket floor boxes, three twin wall sockets and the fixed connection FS01. Each floor-box/socket assembly is individually 13 A fused. Accordingly, the demand calculation uses a 13 A continuous capacity per assembly; FS01 reserves a further 13 A for fixed equipment. Meanwhile, building central AC is supplied as central plant and not duplicated in this office DB.

P1–P4 are A1 rings with fused floor-box connections. In comparison, P5/P6/P8 are A2 radials, while P7 is a fixed radial. Both P6 and P8 serve the PANTRY identified on the source plan, with FB09 and SO01 connected to P6. P8 is a separate socket group for convenient isolation. Kitchen sockets are separated from ordinary office circuits under EMSD6E(3)(a).

Circuit	Points	Phase	Topology	Protection
P1	FB01, FB02	L1	A1 ring	B32
P2	FB03, FB04	L1	A1 ring	B32
P3	FB05, FB06	L2	A1 ring	B32
P4	FB07, FB08	L2	A1 ring	B32
P5	SO02	L1	A2 radial	B32
P6	FB09, SO01	L3	A2 radial	B32
P7	FS01	L3	Fixed radial	B16
P8	SO03	L3	A2 radial	B32

In terms of coverage, the defined service envelopes satisfy the A1 maximum of 100 m^2 and the A2 maximum of 50 m^2. Coverage is assessed by service envelope, allowing overlap between circuits. Ring L/N/CPC return to their own circuit terminals. Main 6/10 mm^2 cables terminate in accessible junction/FCU assemblies rated for those conductor sizes. At each point, the local connections use 4 mm^2 L/N/CPC, with an assumed length of 0.5 m per point.

Task 2 – Layout drawings and MCB board schematics (continued)

FCU and containment details

Every FB01–FB09 assembly shares one 13 A FCU upstream of all four socket ways. Each twin SO01–SO03 likewise shares one 13 A FCU; FS01 has its own switched 13 A FCU. Ring/main L/N/CPC connections must terminate in an accessible junction assembly rated for 6/10 mm^2 cables. At each connection, the FCU is positioned at the tap before the 4 mm^2 local spur, with its neutral switched together with the line and its PE kept continuous. With 13 A continuous capacity per assembly as the demand basis, any added points require recalculation.

To maintain the adopted grouping factors, the containment is arranged in the following banks: lighting LC-A–G (seven circuits, $C_g = 0.54$); A: P1/P2 (four ring legs, 0.65); B: P3/P4 (four ring legs, 0.65); C: P5/P6/P8 (three radials, 0.70); P7 alone (1.00); incoming feeder alone (1.00). Co-located plan paths show routing corridors, not a shared enclosure. Accordingly, separate Method B runs must provide horizontal clear space between groups exceeding twice the largest adjacent overall cable diameter, in accordance with EMSD A5(3), Note 2; where enclosure dimensions govern, the enclosures must be spaced at least as conservatively and the cable separation must be confirmed. Thermal separation is provided by cable clearance, rather than a partition alone. Within each circuit, L/N/CPC stays together. Additionally, contact with thermal insulation must be avoided. If these conditions cannot be achieved, the circuits must be regrouped and resized before installation.

Protection and board arrangement

As shown in the complete schematic, the arrangement includes C63 four-pole protection at MS-O1, 4-core 35 mm^2 plus CPC16, a 100 A four-pole DB isolator, independent L1/L2/L3/N/PE bars and 15 separate 1P+N Type A 30 mA RCBOs. Every load neutral passes through its own RCBO; none are shared downstream. Meanwhile, PE bypasses all switches. Additionally, the circuits and phases are to be labelled, and the arrangement must allow each final circuit to be isolated without interrupting the others.

Device selection must ensure that the proposed B-curve final protection is compatible with driver inrush and the breaking-capacity/energy limits established in Task 3. Individual RCBOs limit the effect of a residual-current fault to its own circuit; however, full short-circuit selectivity requires manufacturer coordination data.

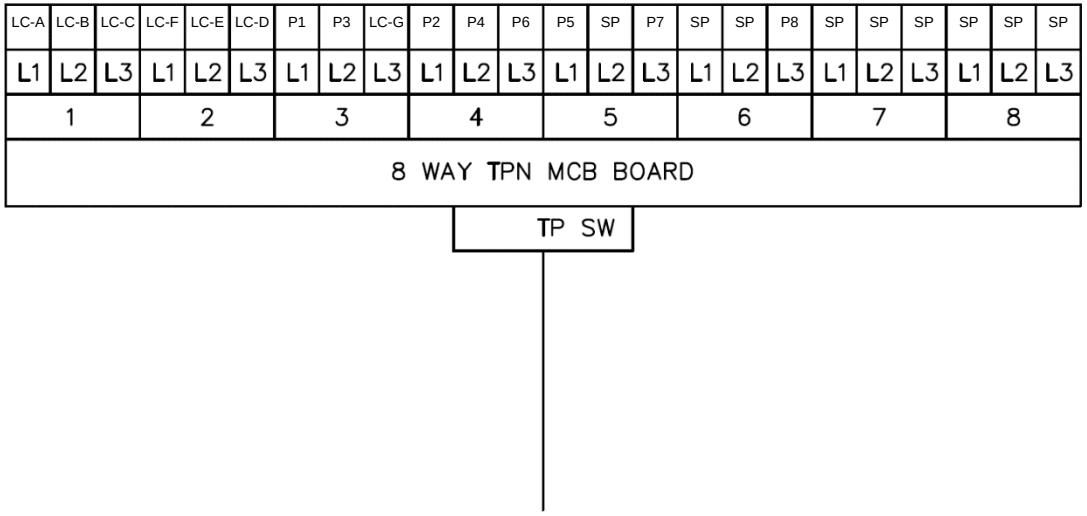
Emergency lighting assumption

Landlord emergency lighting and exit signage use independent essential supplies, separated from normal lighting under EMSD 6B(4). These feeds bypass tenant normal-lighting switches and RCBO load sides. Emergency-system loads are therefore outside this office normal-supply DB.

DB-O1: 8-way TPN chassis; 15 used single-phase outgoing positions

Way	L1	L2	L3
1	LC-A / B10	LC-B / B10	LC-C / B10
2	LC-F / B10	LC-E / B10	LC-D / B10
3	P1 / B32	P3 / B32	LC-G / B10
4	P2 / B32	P4 / B32	P6 / B32
5	P5 / B32	Spare	P7 / B16
6	Spare	Spare	P8 / B32
7	Spare	Spare	Spare
8	Spare	Spare	Spare

Positions indicate phase allocation; each RCBO also needs its own N path. Additionally, the chassis/module capacity for 1P+N devices must be confirmed. The incoming protection is C63 4P at MS-O1, while the local main switch is 100A 4P (including N), supplementing the template's TP SW symbol. The detailed appended schematic governs neutral/PE connections and ring returns. All final RCBOs are Type A, 30mA, 10kA and must meet the energy/inrush specifications.



MCB BOARD SCHEMATIC

Notes
Office 1 DB-O1

15 outgoing circuits
7 lighting / 8 power
Spare ways available
SP = spare

Client
N/A

Project
BSE3124 - Group 1

Drawing
Office 1 distribution board schematic

Date 10 Nov 2026

Project number Group 1

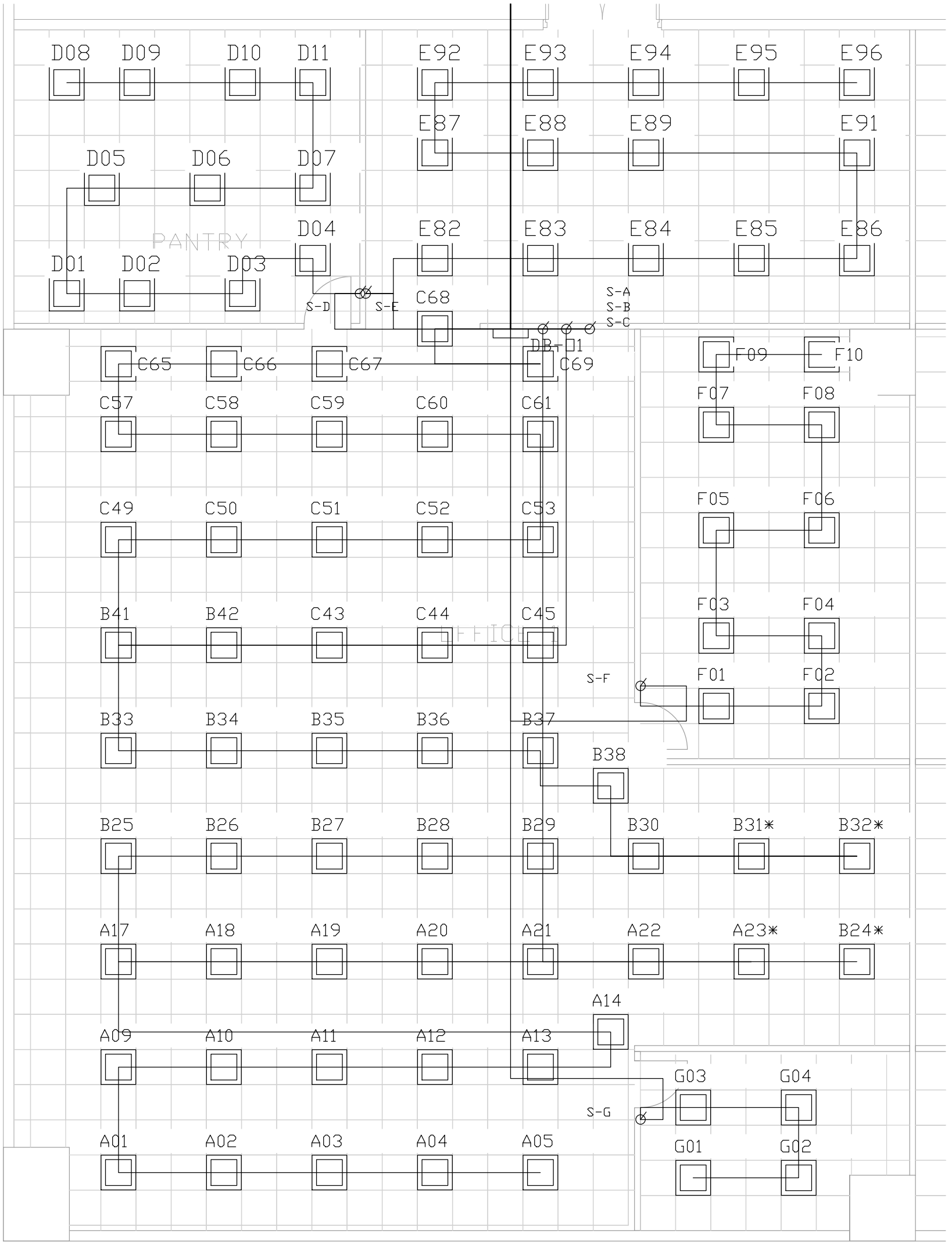
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Scale NTS

TASK 2 | OFFICE 1 NORMAL LIGHTING

A3 PORTRAIT | SCALE 1:75 | DIMENSIONS IN mm | DB-O1

Supply continues to MS-O1 at meter location; incoming route on model / DB schematic.



600x600 LED panel: A/B/C=22 W, *=28 W; D=22 W; E/F=28 W; G=38 W.

Label prefix gives LC-A ... LC-G; suffix retains device ID. S-A ... S-G = local switches.

LC-A L1 18 lamps /402 W B10 2.5 mm2 58.301m
LC-B L2 17 lamps /392 W B10 2.5 mm2 62.184m
LC-C L3 18 lamps /396 W B10 2.5 mm2 44.999m
LC-D L3 11 lamps /242 W B10 2.5 mm2 27.233m
LC-E L2 14 lamps /392 W B10 2.5 mm2 34.464m
LC-F L1 10 lamps /280 W B10 2.5 mm2 32.710m
LC-G L3 4 lamps /152 W B10 2.5 mm2 27.379m

92 panels /2256 W. All circuits: Type A 30 mA1P+N RCBO; L/N/CPC all2.5 mm2.

Manual controls20 AX, LED rating>=600 W; separate phase enclosures. Inrush compatibility required.

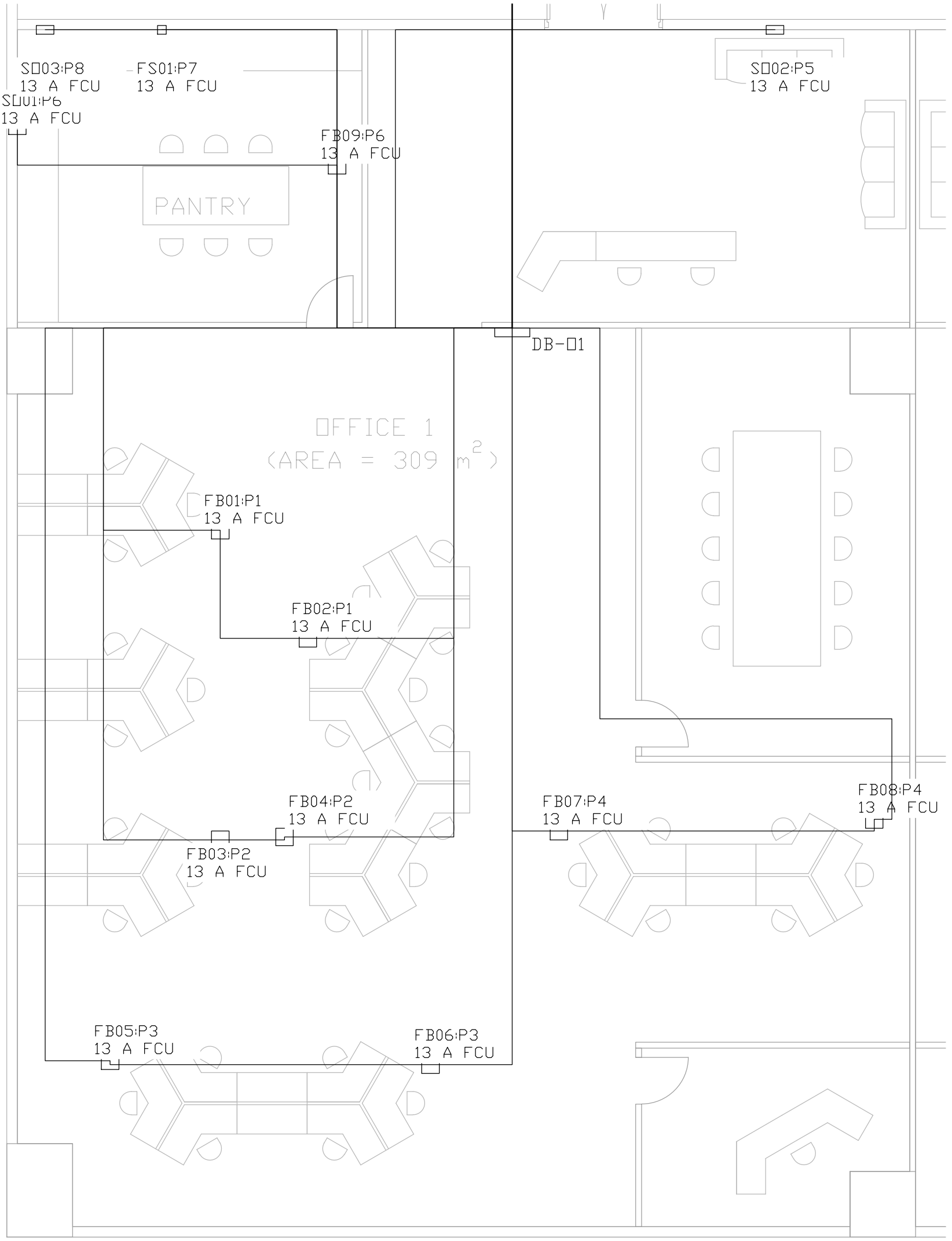
Normal lighting only. Landlord emergency/exit system independently supplied (design assumption).

Proposed routes follow plan; vertical/tail allowances and conditions are in Task3. Architecture retained as reference.

TASK 2 | OFFICE 1 NORMAL POWER

A3 PORTRAIT | SCALE 1:75 | DIMENSIONS IN mm | DB-O1

Supply continues to MS-O1 at meter location; incoming route on model / DB schematic.



FB01-09=floorbox; SO01-03=twin socket; FS01=dedicated fused spur.

Each point:13 A fuse; intended continuous demand<=13 A. Local spur4 mm2,0.5m allowance.

P1 L1 RING B32 6 mm2 28.620m

P2 L1 RING B32 6 mm2 35.520m

P3 L2 RING B32 6 mm2 45.220m

P4 L2 RING B32 6 mm2 34.220m

P5 L1 RADIAL B32 10 mm2 15.607m

P6 L3 RADIAL B32 10 mm2 13.864m

P7 L3 RADIAL B16 4 mm2 13.107m

P8 L3 RADIAL B32 10 mm2 15.107m

Ring L/N/CPC both ends return to same circuit. Radial L/N/CPC same stated size.

FB09/SO01/SO03 serve pantry: P6/P8. P7 is allocated equipment capacity, appliance rating unknown.

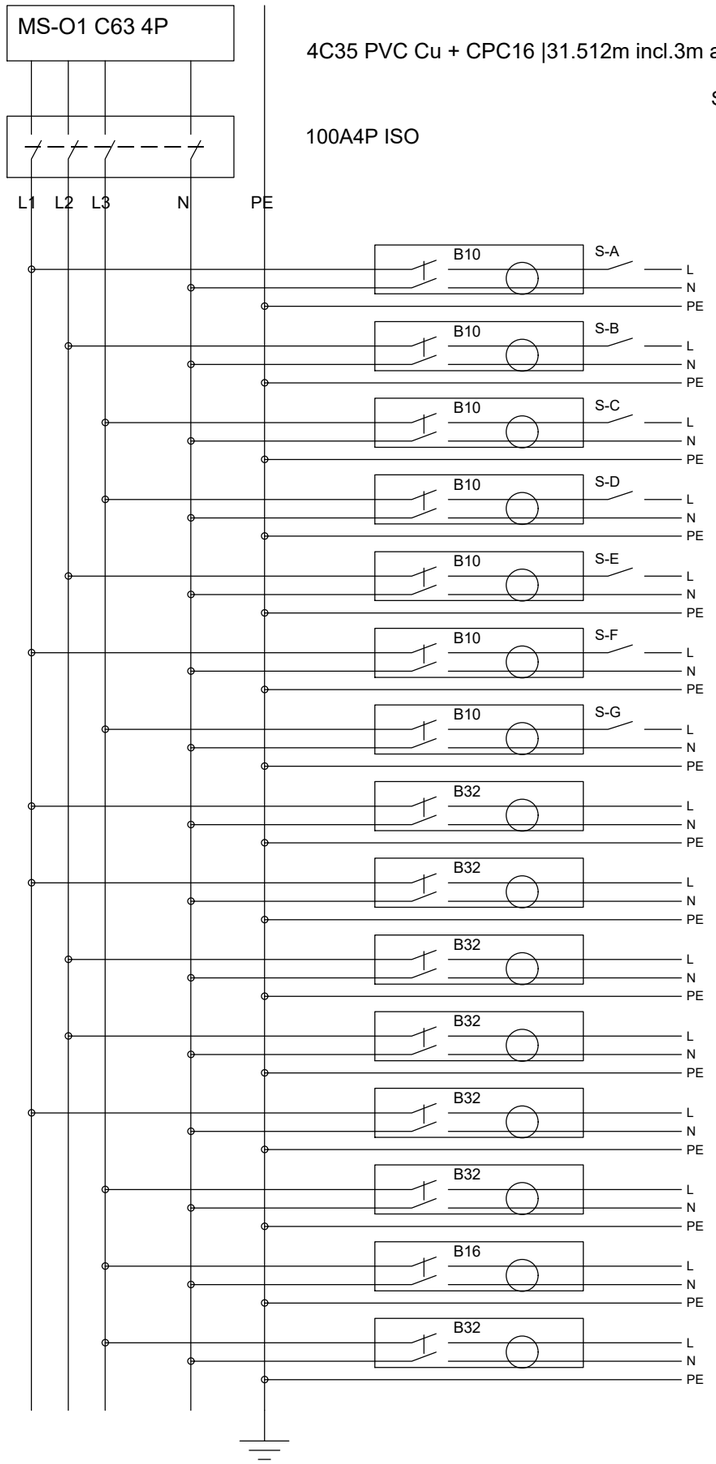
Containment: rings P1/P2 bank A; P3/P4 bank B; radials P5/P6/P8 bank C; P7 separate.

Individually fused point assemblies and terminals must accept selected conductor size or rated junctions.

Proposed routes follow plan; vertical/tail allowances and conditions are in Task3. Architecture retained as reference.

TASK 2 | OFFICE 1 DISTRIBUTION BOARD DB-O1

A3 LANDSCAPE | NOT TO SCALE | DB-O1 | 100 A TPN | 15 OUTGOING CIRCUITS



Source Ze0.20 ohm /PFC<=6 kA assumed; main/RCBO Icn10 kA. See Task3 for conditions and I2t limits.

Landlord emergency/exit systems: independent essential supply assumed; electrically separate from this normal DB.

LC-A [L1] | 18 lamps +S-A /402 W | B10 /TYPE A 30 mA 1P+N RCBO
L/N/PE 2.5/2.5/2.5 mm2 | L=58.301m | LIGHTING RADIAL

LC-B [L2] | 17 lamps +S-B /392 W | B10 /TYPE A 30 mA 1P+N RCBO
L/N/PE 2.5/2.5/2.5 mm2 | L=62.184m | LIGHTING RADIAL

LC-C [L3] | 18 lamps +S-C /396 W | B10 /TYPE A 30 mA 1P+N RCBO
L/N/PE 2.5/2.5/2.5 mm2 | L=44.999m | LIGHTING RADIAL

LC-D [L3] | 11 lamps +S-D /242 W | B10 /TYPE A 30 mA 1P+N RCBO
L/N/PE 2.5/2.5/2.5 mm2 | L=27.233m | LIGHTING RADIAL

LC-E [L2] | 14 lamps +S-E /392 W | B10 /TYPE A 30 mA 1P+N RCBO
L/N/PE 2.5/2.5/2.5 mm2 | L=34.464m | LIGHTING RADIAL

LC-F [L1] | 10 lamps +S-F /280 W | B10 /TYPE A 30 mA 1P+N RCBO
L/N/PE 2.5/2.5/2.5 mm2 | L=32.710m | LIGHTING RADIAL

LC-G [L3] | 4 lamps +S-G /152 W | B10 /TYPE A 30 mA 1P+N RCBO
L/N/PE 2.5/2.5/2.5 mm2 | L=27.379m | LIGHTING RADIAL

P1 [L1] | FB01, FB02 /13 A FUSED POINTS | B32 /TYPE A 30 mA 1P+N RCBO
L/N/PE 6/6/6 mm2 | L=28.620m | A1 RING: L/N/PE BOTH ENDS AT DB

P2 [L1] | FB03, FB04 /13 A FUSED POINTS | B32 /TYPE A 30 mA 1P+N RCBO
L/N/PE 6/6/6 mm2 | L=35.520m | A1 RING: L/N/PE BOTH ENDS AT DB

P3 [L2] | FB05, FB06 /13 A FUSED POINTS | B32 /TYPE A 30 mA 1P+N RCBO
L/N/PE 6/6/6 mm2 | L=45.220m | A1 RING: L/N/PE BOTH ENDS AT DB

P4 [L2] | FB07, FB08 /13 A FUSED POINTS | B32 /TYPE A 30 mA 1P+N RCBO
L/N/PE 6/6/6 mm2 | L=34.220m | A1 RING: L/N/PE BOTH ENDS AT DB

P5 [L1] | SO02 /13 A FUSED POINTS | B32 /TYPE A 30 mA 1P+N RCBO
L/N/PE 10/10/10 mm2 | L=15.607m | DEDICATED RADIAL

P6 [L3] | FB09, SO01 /13 A FUSED POINTS | B32 /TYPE A 30 mA 1P+N RCBO
L/N/PE 10/10/10 mm2 | L=13.864m | DEDICATED RADIAL

P7 [L3] | FS01 /13 A FUSED POINTS | B16 /TYPE A 30 mA 1P+N RCBO
L/N/PE 4/4/4 mm2 | L=13.107m | DEDICATED RADIAL

P8 [L3] | SO03 /13 A FUSED POINTS | B32 /TYPE A 30 mA 1P+N RCBO
L/N/PE 10/10/10 mm2 | L=15.107m | DEDICATED RADIAL

SYMBOLS: box = linked L+N RCBO; circle = residual sensing; dots = junctions; crossings without dots are not connected. PE bypasses all switches.

Each RCBO load neutral is dedicated. Ring L/N/CPC ends return to the same circuit terminals; no mixed neutrals. All local power spurs:13 A fuse +4 mm2 L/N/CPC,0.5m assumed.

S-A/B/C: separate enclosures for different phases. All manual controls20 AX/LED>=600 W minimum; local switching and RCBO inrush performance must meet adopted design specification.

TASK 2 | OFFICE 1 INSTALLATION DETAILS AND DESIGN CONDITIONS

A3 LANDSCAPE | NOT TO SCALE | DIMENSIONS IN mm | OFFICE 1

1. EVERY POWER ASSEMBLY HAS ONE UPSTREAM 13 A FCU

Accessible main/ring junction

13 A switched FCU at tap

4 mm2 L/N/CPC spur
0.5m assumed

ONE assembly: FB / SO / FS
All outlets share 13 A continuous capacity



For rings: both L/N/CPC ends return to their own DB terminals. FCU is before the spur; no unfused extra taps. Main terminals must accept 6/10 mm2.

FB01-09: one FCU per four-socket floorbox. SO01-03: one FCU per twin socket. FS01: one switched FCU. 13 A is not an instantaneous current clamp.

2. SEPARATE CONTAINMENT BANKS - OVERLAPPING PLAN LINES ARE ROUTING CORRIDORS

Lighting LC-A..G	A: P1/P2	B: P3/P4	C: P5/P6/P8	P7 alone	Feeder alone
7 circuits / Cg0.54	4 ring legs / Cg0.65	4 ring legs / Cg0.65	3 radials / Cg0.70	Cg1.00	Cg1.00

Method B, no insulation contact. Keep L/N/CPC of each circuit together. Between banks provide horizontal cable clearance >2x largest adjacent cable OD

(EMSD A5(3) Note2); separate enclosures and verify cable clearance. A partition alone is not thermal separation. If grouping changes, recalculate sizes.

3. A1 RING DESIGN BASIS AND EQUIPMENT SPECIFICATION

EMSD 6C/6E, Table6(1): B32; 6 mm2 copper; corrected leg Iz23.1855 A >=20 A; each service envelope <=100m2; intact L/N/CPC rings; two fused taps per ring.

Ring leg currents14.658-15.203 A support load distribution; overload protection follows the A1 standard arrangement. Verify ring continuity before energisation.

TN-S source Ze0.20 ohm/PFC<=6 kA are design conditions. Main and RCBO Icn10 kA. Required total I2t: B10/B16<=70000; B32/FCU<=90000; C63<=1000000 A2s.

Select apparatus meeting the stated breaking capacity and total clearing energy. Control20 AX/LED>=600 W; confirm driver inrush. PE bypasses switching.

4. LIGHTING DESIGN BASIS

92 panels /2256 W. LDT calculation, MF0.80, luminaires1.9m above workplane: mean direct lux open592.9 /pantry546.3 /upper east588.5 /south east662.6.

South east: four38 W panels,4685.55 lm each; mean589.2 lx at2.2m above workplane,17.8% above500 lx target. Lighting positions and routes retained.

Adopted height1.9m and MF0.80 with periodic cleaning. Direct-light calculation; see report for room geometry and perimeter illuminance.

Task 3 – Maximum current demand calculation and cable sizing

Maximum current demand calculation:

Lighting current is calculated as $I = P/(220 \times 0.90)$. Under EMSD Table 7(1), the demand includes 90% of lighting and 100% of the largest Code 6E circuit plus 50% of the remaining circuits on each phase. P7 is allocated separately at 100%. The 13 A per assembly is an intended maximum continuous load, not an instantaneous fuse clamp.

Phase	Light A*	Sockets A	Fixed A	Demand A
L1	3.444	45.500	0	48.600
L2	3.960	39.000	0	42.564
L3	3.990	32.500	13	49.091

*Lighting column is before the 90% factor. Thus, the maximum demand of **49.091 A** leads to the selection of 63 A source protection and a 100 A DB. Maximum undiversified phase allowance is 68.444 A. Full branch currents are retained for outgoing cable sizing.

Cable sizing for the incoming circuit to the distribution board:

The selected feeder is 4-core 35 mm^2 PVC copper plus 16 mm^2 CPC, using Method B. Its 28.5119 m route plus an assumed 3 m vertical/termination allowance totals 31.5119 m. At 40 °C, $C_a = 0.87$, $C_g = 1$; the calculation adopts $C_h = 0.86$ for the stated third-harmonic range (15–33%).

$$
\begin{aligned}
I_{z,25} &= 80(0.87)(0.86) = 59.856 < 63 \text{ A}, \\
I_{z,35} &= 99(0.87)(0.86) = 74.0718 \text{ A}, \\
I_b &= 49.091 \le I_n = 63 \le I_z = 74.0718 \text{ A}.
\end{aligned}
$$

These ratings follow EMSD A6(2), column 5, while CPC sizing follows Table 11(2)(a). A full-size neutral is retained; neutral current must remain within the adopted bound.

At 63 A, feeder drop is 2.521606 V. Including the 2.2 V pre-origin allowance and calculated branch drops gives a worst complete drop of 3.031% of 220 V, below 4%. The checks assume TN-S, source $Z_e = 0.20\,\Omega$ and PFC ≤ 6 kA. Thus, the DB external loop is 0.264129 Ω; source C63 disconnection and outgoing B-curve checks pass under those conditions.

Outgoing circuits are also included, as required by the assignment: all 15 sizes, currents, lengths, voltage drops, earth-fault checks and apparatus energy requirements are tabulated in the continuation.

Task 3 – Maximum current demand calculation and cable sizing (continued)

Incoming design conditions

The preceding sheet selects C63 protection and a 4-core 35 mm^2 feeder with 16 mm^2 CPC. Along the meter-to-DB route, the length is 28.5119 m plus a 3 m vertical/termination allowance. Cable sizing assumes 40 °C, a separate Method B enclosure and third-harmonic content greater than 15% and at most 33%, giving $C_a = 0.87$, $C_g = 1$ and $C_h = 0.86$. A full-size neutral is retained on this basis.

To bound the phase-neutral voltage drop, the calculation uses $(2/\sqrt{3})(1.10) = 1.270170592$ mV/A/m from the three-phase table impedance, with neutral current at most 63 A. Thus, feeder drop is 2.521606 V at 63 A and 1.964888 V at the design demand of 49.091 A. Additionally, the complete-path calculation below includes the 2.2 V upstream allowance.

All outgoing cables

All outgoing cable sizes refer to PVC copper L/N/CPC conductors. Lighting grouped separately has $I_z = 24(0.87)(0.54) = 11.2752$ A for seven 2.5 mm^2 circuits; all $I_b \leq 10 \leq I_z$. Lighting lengths include 5 m DB/switch vertical allowance.

Each uniform ring bank counts as four current-carrying legs, giving $I_z = 41(0.87)(0.65) = 23.1855$ A per 6 mm^2 leg, above the standard 20 A minimum. With 13 A per point, the ring-network solution gives maximum leg currents 14.658–15.203 A, below I_z. However, the radial criterion $I_n \leq I_z$ must not be applied to just one ring leg. Continuity and fused-point restrictions are essential.

Meanwhile, the 10 mm^2 A2 radial cables have $I_z = 57(0.87)(0.70) = 34.713$ A; P7's separate 4 mm^2 radial has $I_z = 32(0.87) = 27.84$ A. In determining route lengths, the power radials include a 2 m vertical allowance, while the rings include 2 m at each DB end. Additionally, each local point spur adds 0.5 m of 4 mm^2 cable.

Circuit	Phase	I_b A	RCBO A	L/N/CPC mm^2	Main length m
LC-A	L1	2.030	10	2.5	58.301
LC-B	L2	1.980	10	2.5	62.184
LC-C	L3	2.000	10	2.5	44.999
LC-D	L3	1.222	10	2.5	27.233
LC-E	L2	1.980	10	2.5	34.464
LC-F	L1	1.414	10	2.5	32.710
LC-G	L3	0.768	10	2.5	27.379
P1	L1	26.000	32	6	28.620
P2	L1	26.000	32	6	35.520
P3	L2	26.000	32	6	45.220
P4	L2	26.000	32	6	34.220
P5	L1	13.000	32	10	15.607
P6	L3	26.000	32	10	13.864
P7	L3	13.000	16	4	13.107
P8	L3	13.000	32	10	15.107

Task 3 – Maximum current demand calculation and cable sizing (continued)

Complete voltage drop and fault checks

The voltage-drop calculation sums the segment drops using downstream currents on the radials and solves the two-feed ring network, including the stated vertical/local allowances. Combining this result with the 2.2 V pre-origin allowance and the 2.521606 V feeder bound gives the full-path drop, which is checked against a limit of $0.04(220) = 8.8$ V.

Fault-protection checks assume a TN-S supply, an external source loop impedance of 0.20 ohm and PFC at most 6 kA at MS-O1/downstream. Adding the incoming phase+CPC bound of 0.064129 ohm gives a DB external loop impedance of 0.264129 ohm. Table 11(10) reference limits are B10=4.17, B16=2.61, B32=1.31 and C63=0.33 ohm. Throughout these checks, the calculation uses operating-temperature paths and a minimum voltage factor of 0.95. Within a uniform ring, the worst parallel path is at most a quarter of the circumference.

Circuit	Out drop V	Total drop %	Z_s ohm	Min fault A	Z_s limit ohm
LC-A	1.450	2.805	1.314	159.1	4.17
LC-B	1.237	2.709	1.383	151.1	4.17
LC-C	0.959	2.582	1.074	194.6	4.17
LC-D	0.422	2.338	0.754	277.1	4.17
LC-E	0.797	2.509	0.884	236.3	4.17
LC-F	0.642	2.438	0.853	245.0	4.17
LC-G	0.345	2.303	0.757	276.1	4.17
P1	1.269	2.723	0.322	649.3	1.31
P2	1.675	2.907	0.334	624.9	1.31
P3	1.937	3.027	0.352	593.5	1.31
P4	1.428	2.795	0.332	629.4	1.31
P5	0.964	2.584	0.338	617.8	1.31
P6	1.310	2.742	0.331	632.1	1.31
P7	1.946	3.031	0.414	505.1	2.61
P8	0.936	2.571	0.336	621.8	1.31

Under the adopted conditions, all 15 full voltage bounds pass, with a worst-case drop of 3.031%. Additionally, the calculated minimum earth-fault current in each B-curve final circuit exceeds the threshold $5I_n$. Similarly, incoming C63 gives $Z_s = 0.264129 < 0.33$ ohm and $I_{f,\min} = 791.28 > 10(63) = 630$ A, satisfying reference instantaneous disconnection under the adopted conditions.

Task 3 – Maximum current demand calculation and cable sizing (continued)

Overload and total clearing energy

CPC thermal withstand is checked using $k = 115$ for bunched PVC copper CPCs initially at 70 °C (11(2)(c)), with total $I^2t \leq (kS)^2$ required. Additionally, the main protection and RCBOs must have a breaking capacity of 10 kA. B-curve devices must limit class 3-equivalent total energy to at most 70,000 A^2s for B10/B16 and 90,000 for B32. Accordingly, the selected RCBOs must have manufacturer total-clearing-energy data that meet these limits. Corresponding CPC withstands are 82,656.25 (2.5 mm^2), 211,600 (4), 476,100 (6) and 1,322,500 (10), each above the required device bound.

Incoming C63 total energy must be at most 1,000,000 A^2s over the relevant fault range, below CPC16 withstand 3,385,600. Similarly, complete 13 A FCU energy must be at most 90,000, below the 4 mm^2 spur withstand 211,600. Therefore, the incoming breaker and FCU assemblies must be selected to meet these total-energy limits.

Under fuse overload conditions, $I_2 = 1.9(13) = 24.7$ A. Spur $I_z = 32(0.87)(0.70) = 19.488$ A gives $1.45I_z = 28.2576 > 24.7$ A. Overload suitability of P1–P4 is assessed using the standard A1 arrangement in EMSD 6C, 6E and Table 6(1), Note 1(i): 32 A protection, copper size at least 2.5 mm^2, corrected per-leg capacity at least 20 A, served area at most 100 m^2, intact L/N/CPC rings and evenly distributed outlets. On this basis, the adopted 6 mm^2 rings meet these conditions. Exactly two individually 13 A fused connections are allowed per ring; no additional unfused taps are permitted. Here, the leg-current calculation checks load distribution; the A1 standard arrangement governs ring overload protection. With a reference interruption rating of 6 kA, the fuse meets the assumed PFC of at most 6 kA. However, its typical pre-arcing 870 A^2s is *not* total clearing energy. Finally, the design assumes an FCU ambient temperature of at most 35 °C and a board apparatus ambient temperature of at most 30 °C; a containment temperature of 40 °C remains the conservative cable condition.

References

CLP COP215 Rev 07: clauses 3.1.7 and 3.3.1; Appendices 2 and 7 (building demand). CLP COP101 v16: clauses 4.3, 5.5, 5.6 and 6.1–6.7 (plantroom planning).

EMSD Code of Practice for the Electricity (Wiring) Regulations, 2025: Table 6(1), printed p.56; Table 7(1), pp.65–67; 9B, p.79; Table 11(2), pp.99–100; Table 11(10), p.105; Appendix 5 correction factors and Appendix 6 cable ratings.

Performance in Lighting, PLATO 600 MultiPower 3117955: LOW (22 W), MEDIUMLOW (28 W) and HIGH (38 W) manufacturer LDT files, October 2022. ABB guide 9AKK108468A3912, p.130: B-curve energy references. Eaton TDC180 Technical Data 2042, August 2024, p.2: fuse conventional-fusing and interruption data. Total FCU clearing energy is specified separately above.

Electrical Schematic 1: transformer, main LV switchboard and floor-metering arrangement. Electrical Schematic 2: final distribution-board schematic conventions.